WORKED SOLUTION

STEP III 2006, Question 10

Mechanics • Conservation of Angular Momentum

STEP III 2006, Question 10

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/06-s3-q10

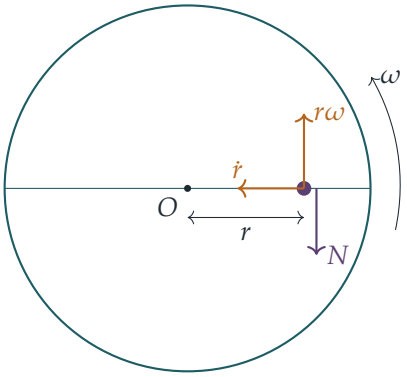
The key idea

Nothing external acts on the disc-and-particle system to turn it about the axis, and the smooth groove does no net work, so there are two conservation laws: angular momentum about the axis and kinetic energy. These give ω and $\dot{r}$ as functions of r alone, and the strange-looking condition linking V, Ω, a and k is there to make the energy expression collapse. Dividing $\dot{r}$ by $\dot{\theta} = \omega$ removes time, which leaves a separable equation for the path.

Solution

The model

We treat the particle as a point mass m, and the groove as smooth, so the only horizontal force on the particle is a reaction N perpendicular to the groove. The disc turns freely about a fixed vertical axis, so the bearing exerts no torque about it. Gravity and the vertical reactions are vertical and play no part. Let θ be the angle the disc has turned through, so $\omega = \dot{\theta}$, and let r be the particle's distance from the axis. Because the particle stays in the groove, its velocity has a component $\dot{r}$ along the groove and a component $r\omega$ perpendicular to it.



Angular speed

The reaction N acts on the particle and $-N$ acts on the disc at the same point, so their moments about the axis cancel. With no external torque about the axis, the total angular momentum about it is constant. The disc contributes $mk^2\omega$, and the particle contributes $mr \cdot r\omega = mr^2\omega$ (only the transverse component of its velocity has a moment). At $t = 0$ the total is $m(k^2 + a^2)\Omega$, so

$$(k^2 + r^2)\,\omega = (k^2 + a^2)\,\Omega, \qquad \text{that is,} \qquad \omega = \frac{\Omega(k^2 + a^2)}{k^2 + r^2},$$

which is the first result.

Radial speed

Ask yourself The groove pushes the particle sideways while the particle moves. Why is energy conserved?

The reaction N is perpendicular to the groove. The particle and the point of the disc in contact with it have the same velocity component perpendicular to the groove, namely $r\omega$; they differ only by the sliding velocity $\dot{r}$ along the groove, which is perpendicular to N. So the work N does on the particle is cancelled exactly by the work $-N$ does on the disc. The bearing at the axis does no work, so the total kinetic energy is constant.

Cancelling the common factor $\frac{1}{2}m$, conservation of energy gives

$$k^2\omega^2 + \dot{r}^2 + r^2\omega^2 = k^2\Omega^2 + V^2 + a^2\Omega^2.$$

Now we use the given condition $V^2 = \Omega^2 a^2(k^2 + a^2)/k^2$ on the right:

$$(k^2 + a^2)\Omega^2 + \frac{\Omega^2 a^2(k^2 + a^2)}{k^2} = \frac{\Omega^2(k^2 + a^2)(k^2 + a^2)}{k^2} = \frac{\Omega^2(k^2 + a^2)^2}{k^2}.$$

On the left, $(k^2+r^2)\omega^2 = \Omega^2(k^2+a^2)^2/(k^2+r^2)$ by the first result. Hence

$$\begin{aligned}\dot{r}^2 &= \Omega^2(k^2+a^2)^2\left(\frac{1}{k^2}-\frac{1}{k^2+r^2}\right)\\ &= \frac{\Omega^2 r^2(k^2+a^2)^2}{k^2(k^2+r^2)},\end{aligned}$$

as required.

The equation of the path

While the particle moves towards the axis, $\dot{r} < 0$, so taking the negative square root,

$$\dot{r} = -\frac{\Omega r(k^2+a^2)}{k\sqrt{k^2+r^2}}.$$

Since $\omega = \dot{\theta} > 0$, we can use the chain rule, $\dfrac{\mathrm{d}r}{\mathrm{d}\theta} = \dot{r}\Big/\dot{\theta} = \dot{r}/\omega$:

$$\frac{\mathrm{d}r}{\mathrm{d}\theta} = -\frac{\Omega r(k^2+a^2)}{k\sqrt{k^2+r^2}}\cdot\frac{k^2+r^2}{\Omega(k^2+a^2)} = -\frac{r\sqrt{k^2+r^2}}{k},$$

so $k\dfrac{\mathrm{d}r}{\mathrm{d}\theta} = -r(k^2+r^2)^{1/2}$, as required.

Solving for r

Put $u = k/r$, so $u > 0$. Then

$$\frac{\mathrm{d}u}{\mathrm{d}\theta} = -\frac{k}{r^2}\frac{\mathrm{d}r}{\mathrm{d}\theta} = \frac{k}{r^2}\cdot\frac{r\sqrt{k^2+r^2}}{k} = \sqrt{\frac{k^2}{r^2}+1} = \sqrt{1+u^2}.$$

This separates:

$$\int\frac{\mathrm{d}u}{\sqrt{1+u^2}} = \int\mathrm{d}\theta \quad\Longrightarrow\quad \operatorname{arsinh} u = \theta+\alpha$$

for some constant α. So $k/r = \sinh(\theta+\alpha)$, that is,

$$r\sinh(\theta+\alpha) = k.$$

At $t = 0$ we have $\theta = 0$ and $r = a$, so $\sinh\alpha = k/a$, as required.

The particle never reaches the axis

> **Ask yourself** $r = k/\sinh(\theta+\alpha)$ is never zero, but could θ become infinite in a finite time?

No. Since $r^2 \geq 0$, the first result gives $\omega \leq \Omega(k^2+a^2)/k^2$, so $\theta \leq \Omega(k^2+a^2)t/k^2$ is finite at every finite time t. Then $r = k/\sinh(\theta+\alpha) > 0$ at every time.

We should also check that the formulae stay valid, that is, that the particle keeps moving towards the axis. By the formula for $\dot{r}^2$, $\dot{r}$ can only vanish when $r = 0$. So while $r > 0$, $\dot{r}$ keeps its initial negative sign, and the argument above applies for all $t > 0$. The particle spirals ever closer to the axis, with $r \to 0$ only as $\theta \to \infty$, but it never reaches it.

Checks

- At $r = a$ the formula for $\dot{r}^2$ gives $\Omega^2 a^2(k^2 + a^2)/k^2 = V^2$, matching the initial condition.
- With $k = a = \Omega = 1$ (so $V = \sqrt{2}$), we integrated the equations of motion $\ddot{r} = r\omega^2$ and $(k^2 + r^2)\omega =$ constant numerically. The particle reached $r = 0.01$ after the disc had turned through $\theta \approx 4.4170$, and $\operatorname{arsinh} 100 - \operatorname{arsinh} 1 \approx 5.2983 - 0.8814 = 4.4169$ agrees. Energy, $\dot{r}^2$ and $r\sinh(\theta + \alpha) = k$ all held along the run to about 10^{-11}.

Common slips

- Forgetting the particle's own angular momentum $mr^2\omega$, or its transverse kinetic energy $\frac{1}{2}mr^2\omega^2$.
- Taking the positive square root for $\dot{r}$, which gives the wrong sign in $\mathrm{d}r/\mathrm{d}\theta$.
- Concluding "never reaches the axis" from $r \neq 0$ alone, without noting that θ stays finite at finite times.

Beyond STEP

The particle feels no radial force, yet in the rotating frame of the disc it seems to be pushed outwards. That is the centrifugal term in $\ddot{r} = r\omega^2$, and the sideways reaction N balances the Coriolis force. Undergraduate mechanics handles problems like this with Lagrangian methods: the angle θ does not appear in the energy, so its conjugate momentum, here the angular momentum, is conserved automatically. Kleppner and Kolenkow, *An Introduction to Mechanics*, covers rotating frames and angular momentum at the right level.

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